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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (VDE) (CBCS) Degree Examinations

December 2024 - January 2025
22VDE112 – VLSI DESIGN VERIFICATION

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each
and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	VLSI technology trends affect the testing. Justify the statement with appropriate explanation.	10	L*2	1	3
b)	With the help of a neat design flow, discuss the SOC verification technology.	10	L2	1	3
2. a)	With a neat flow chart, explain in detail the VLSI development process.	10	L2	1	3
b)	Discuss the formal model checking and formal equivalence checking in detail.	10	L2	2	3
3. a)	What are the factors to be considered while creating a system level test bench? Explain in detail.	10	L2	2	3
b)	What are IP blocks? Explain different types of IP blocks available for functional verification.	10	L2	1	3
Unit – II					
4. a)	What is cross talk? What are the reasons for cross talk? How does it create problems for signal integrity?	10	L2	4	3
b)	What is parasitic extraction? Discuss in detail the different methodologies of parasitic extraction.	10	L2	4	3
5. a)	Explain in detail PVT variation and analysis corner.	10	L2	4	3
b)	Describe in detail Design Rules check and Electrical Rules Check.	10	L2	4	3
6. a)	IR drop and electro migration effects create problems signal integrity. Justify.	10	L2	4	3
b)	Explain the signal Integrity Analysis steps with neat diagram.	10	L2	4	3
Unit – III					
7. a)	Explain the Scan design process and Scan Design Rules.	10	L2	5	3
b)	Characterization test and production tests are the two names related to VLSI chip testing. Briefly explain them with relative comparison.	10	L2	5	3
8. a)	Explain the following electrical parametric tests.(1)Threshold Test (2) Set up, Hold and Release Time Test (4) Propagation Delay Test.	10	L2	5	3
b)	What is the importance of testing in VLSI process? Explain the principle of testing in VLSI.	10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
